



Red-eared Slider from Indian Sunderban; a Rising Threat over Indigenous Biodiversity

J. S. Yogesh Kumar¹ · Arya Sen¹ · Prabhakar Rai¹ · Kaushik Deuti²

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Abstract

The red-eared slider turtle, a highly sought-after pet globally, has become one of the most problematic invasive species, originating from the Mississippi River basin in North America. With millions exported worldwide, it has invaded numerous countries, posing threats to native ecosystems due to its aggressive behaviour and competition. Recent observations in India indicate its presence in natural habitats, raising concerns. In February 2024, a live red-eared slider was discovered near the Sunderban Biosphere Reserve, marking the first known instance in an estuarine environment, highlighting its adaptability. While its effects are well-documented in many countries, research on this issue in India is somewhat limited. Native turtle species, particularly those in the Gangetic basin, face threats from various factors, including the growing ornamental pet trade, which worsens the situation due to consumer ignorance. Implementing effective control measures is crucial for maintaining sustainable biodiversity and preventing the spread of red-eared sliders.

Keywords Chelonians · Invasive species · Ornamental pet · Sunderban Biosphere Reserve

The red-eared slider (RES) turtle, globally renowned as a highly sought-after pet, is also recognized as one of the most problematic invasive species worldwide and has a stable population in Bermuda, Canada, Australia, Europe, Great Britain, South Africa, Caribbean Islands, Israel, Bahrain, Mariana Islands, Guam, Russia, south-east Asia [1].

Originating mainly from the Mississippi River basin in North America, it has been exported in vast numbers, with around 52 million individuals sent from the United States to other countries between 1989 and 1997 [1]. This species has now been introduced to at least 72 countries and has established itself as an invasive species in 22 of them [1]. Studies have extensively documented its impact, revealing its aggressive behaviour, competition, and predation wreak havoc on native ecosystems [2]. Additionally, there are concerns about the transmission of infectious diseases to humans [2]. In India, numerous observations of red-eared sliders have been recorded, with more reports emerging through citizen science platforms documenting their presence in natural aquatic habitats [2].

Recently, on February 23, 2024, during a routine faunal survey near Dabu, Canning (22°16'27.10"N, 88°39'46.14"E), a RES (Total length – 6.4 cm) was recorded (Fig. 1). The turtle was found in the possession of a fisherman who was using cast netting to catch fish in a small channel connected to the Malta River. Following its capture, the live specimen underwent morphological identification and was subsequently preserved in the National Zoological Collection of the Zoological Survey of India (Registration No KN9403). 16s rRNA from the specimen was extracted, amplified, sequenced following Sreeraj et al. [3] and the sequence was checked using NCBI BLAST, deposited in NCBI (Submission ID SUB14664295). This marks the first documented instance of a red-eared slider being found on the outskirts of the Sunderban Biosphere Reserve, and notably, it is the first time a live specimen has been captured from an estuarine environment, also the sequence deposited in NCBI is first from Indian region. Further examination of the channel revealed a salinity level of 8 ppm, indicating that these turtles are capable of surviving in estuarine conditions.

16s rRNA phylogenetic analysis was performed using 7 NCBI database *Trachemys scripta* 16s rRNA sequences (Accession Numbers: MW122291.1, MN135556.1,

✉ J. S. Yogesh Kumar
yogeshkumarzsi16@gmail.com

¹ Zoological Survey of India, Sunderban Regional Centre, Canning, West Bengal 743329, India

² Herpetology Division, Zoological Survey of India, 27 JL Nehru Road, Kolkata, West Bengal 700016, India



Fig. 1 Collected *trachemys scripta elegans* from the study site

L28077.1, KM216749.1, KM216748.1, AB090050.1, MT137277.1) and one current study sequences. The study uses 1634-character alignments. The group has 299 parsimony informative, 1061 variable, and 559 conserved sites. NJ analysis of sequencing data produced a phylogenetic tree with two main clades and an outgroup (Fig. 2). This phylogenetic analysis clarified the relationships among *T. scripta* from different regions, identifying two distinct clades and one outgroup in the phylogenetic tree. Clade 1 includes *T. scripta* exclusively from South Korea, China, Japan, and

the USA, while Clade 2 consists only of *T. scripta* from the USA. The significant phylogenetic similarity within Clade 1 may be due to shared habitats and environmental conditions, particularly in areas where the current generation of *T. scripta*, imported from the USA, has adapted, distinguishing them from the native population found in Clade 2 [4]. The outgroup specimen (present study specimen) likely appears separate due to its origin in an estuarine ecosystem, which differs significantly from the native habitat, emphasizing the role of habitat in phylogenetic diversification [5].

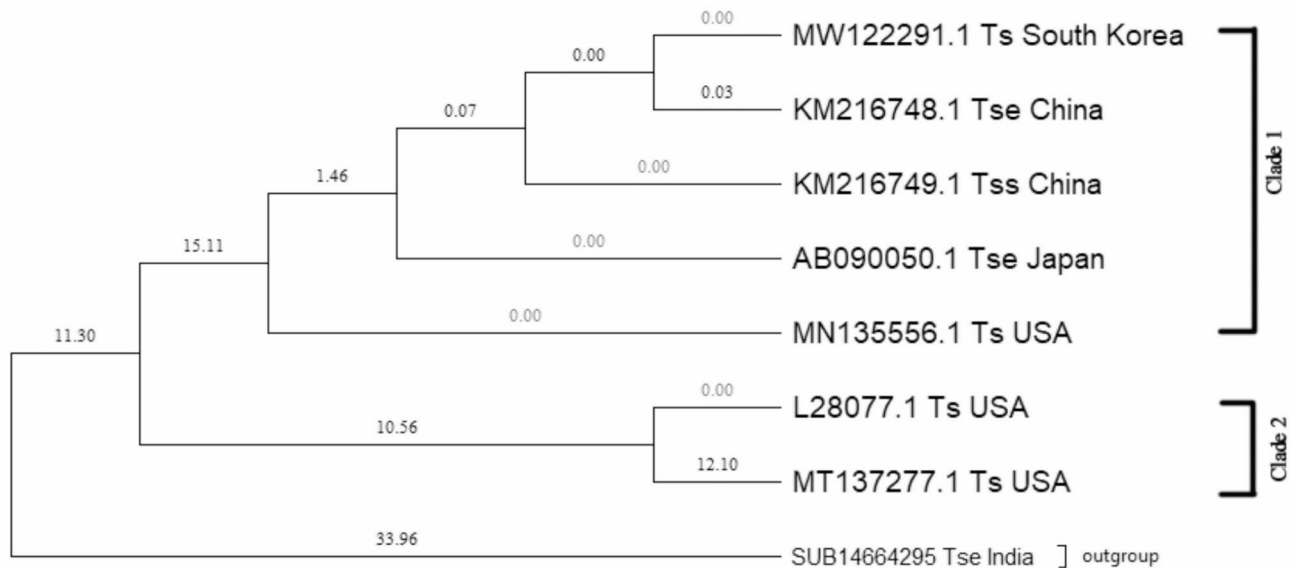


Fig. 2 Phylogenetic relationships of *trachemys scripta* of present study with other *trachemys scripta* (available in NCBI); derived by NJ analysis of the 16s rRNA sequences

While the ecological impacts of RES on native biodiversity are well-documented in places like Europe and Australia, our understanding remains limited in tropical regions [4]. RES possess competitive advantages over native and endemic turtle species due to their early maturation, aggression, higher fecundity, and large adult body size. Research from Europe demonstrates that RES have adversely affected native and threatened turtle species by competing for food resources [6]. Additionally, RES are known to prey on waterbird eggs, and hatchlings, and occupy nesting sites. Their omnivorous diet also raises concern about their impact on various aquatic prey species, including rare amphibians [7]. Consequently, the implications of RES extends beyond simple competition, and their presence in non-native habitats demands special attention to mitigate potential wider ecological and economic losses.

India hosts a rich variety of native freshwater turtles, totaling 22 species, among which 16 are under threat, including those classified as Critically Endangered, Endangered, and Vulnerable, while three are Near Threatened, and three are Endemic [2]. In the Sunderban, India's largest delta, 12 species of marine and estuarine turtles are present, receiving protection due to their low rates of survival and breeding. Also, the Gangetic basin boasts a significant diversity of freshwater turtles but is identified as a threatened extinction zone for the Chelonians [8]. In the northern section of the Hooghly River, environmental conditions are conducive for RES. These areas directly connect to the Bay of Bengal in the south and various small channels in the Gangetic basin utilized for irrigation. Consequently, these

channels provide optimal pathways for RES migration into extensive areas of West Bengal and the Sunderban, posing a substantial future threat to the native biodiversity. RES has already been documented from Gujarat, Karnataka, Andhra Pradesh, Punjab, West Bengal, Rajasthan, Goa and north-eastern states of India (Fig. 3) [9].

The growing ornamental pet industry significantly contributes to the current challenges, driven by factors such as consumer ignorance, the ease of maintaining RES, and their high reproductive rates. RES experiences rapid initial growth but can become less appealing and harder to manage in small aquariums once they reach a certain size [10]. As a result, many consumers release them into local water bodies, threatening natural biodiversity. Similar patterns are seen with aquarium crocodile fish. This issue is further compounded by illegal trading fueled by high market demand. Currently, India lacks specific laws or robust policies addressing invasive alien species [11]. A notable incident in 2013 involved the seizure of about 10,000 RES hatchlings being smuggled into India at the Kolkata International Airport, highlighting the scale of illegal trade [12]. RES pose an invasion risk to natural water bodies across northeast India, which are home to 21 of the 29 vulnerable native freshwater turtle species. Between August 2018 and June 2019, herpetologists from the NGO "Help Earth" found RES in the Deepor Beel wildlife sanctuary and the Ugratara Devalaya temple pond, with additional reports of their presence in a stream feeding into the Tawng river and on a farm in Aizawl, Mizoram. This indicates that consumer ignorance is a key driver of the problem. The species'



Fig. 3 Known distributional records and present record of *Trachemys scripta* from India

potential distribution is expected to increase significantly with future climate change, and the ongoing introduction into the wild is likely to continue and even intensify as trade and availability grow [13, 14]. Therefore, effective control measures are essential to prevent further worsening of the situation.

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Author contributions Author JSYK collected the specimen, confirmed the identification and prepared the manuscript. Authors AS, PR assisted in the survey and identification. Author KD verified the identification and assisted in MS preparation.

Significance Statement The MS describes red-eared slider turtle recorded estuarine habitat of, Sunderban Biosphere Reserve. Such work from a biosphere reserve needs documentation for future conservation efforts and protection against bio-invasion.

Competing Interests The authors have no competing interests to declare that are relevant to the content of this article.

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